MACHINE LEARNING AND DATA MINING · (MODULE I) FINAL ASSIGNMENT

Predicting Hotel Booking Cancellations

A CRISP-DM approach on 119,390 real bookings

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1 · BUSINESS UNDERSTANDING

An empty room-night is revenue lost forever



Goal

Predict whether a booking will be cancelled, using only information available **at booking time**, and find the customers most at risk.



Actions supported

Low-cost actions: **contact** high-risk guests before arrival and **ask for a deposit** on the riskiest bookings.



Cost of errors

False negative

missed cancellation → empty room

False positive

false alarm → one extra reminder

→ **Recall first**, plus F1 and PR-AUC

2 · DATA UNDERSTANDING

Real bookings from two Portuguese hotels

119,390
bookings, 2015–2017

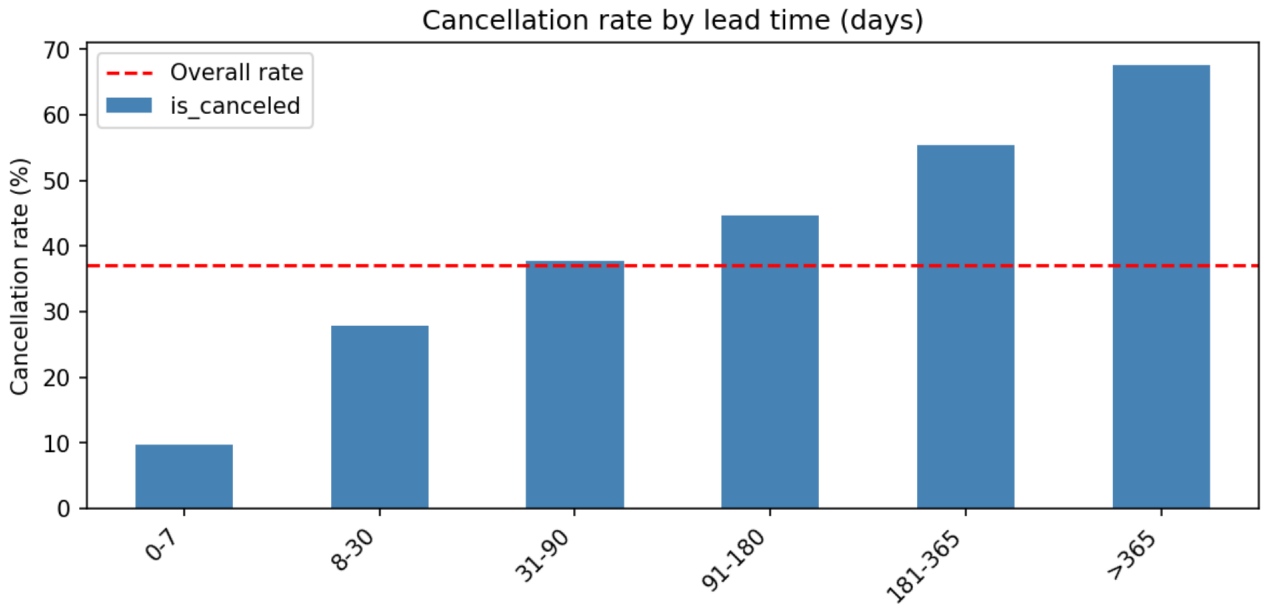
32
attributes: booking, channel, customer, rate

37%
of bookings cancelled (target **is_canceled**)

City Hotel and Resort Hotel · Source: Antonio, Almeida & Nunes, *Data in Brief*, 2019. The data need real preparation: missing values, duplicates, anomalous rows and hidden leakage.

2 · EXPLORATORY ANALYSIS

The earlier the booking, the higher the risk



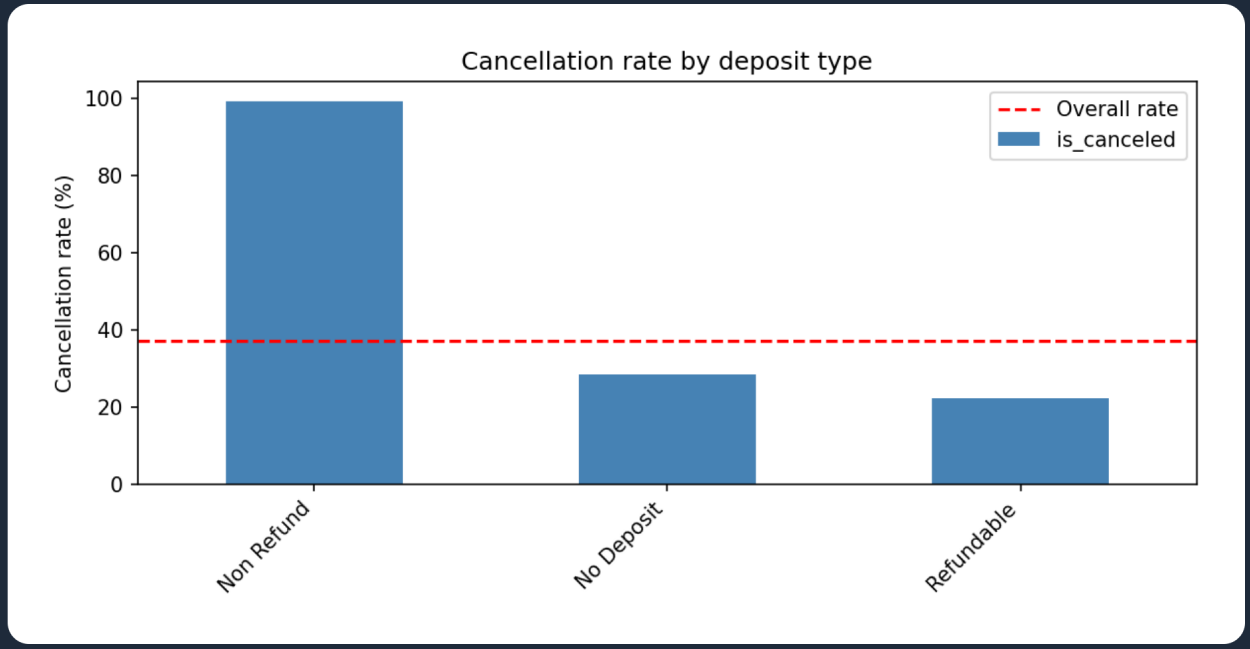
9.6% → 67.7%

cancelled within 7 days vs over 1 year ahead

City Hotel cancels more (41.7% vs 27.8%), as do Groups and online agencies. **Special requests** lower the risk.

2 · THE NON REFUND PARADOX

99% of non-refundable bookings were cancelled. Or were they?



A non-refundable deposit should *discourage* cancellations. These bookings were almost all City Hotel, Portuguese, Groups or tour operators.

14,587 → 1,037

Non Refund bookings after removing duplicates: mostly **repeated group blocks**

2 · HIDDEN DATA LEAKAGE

Some attributes already know the answer

reservation_status

= outcome

Says “Canceled” or “Check-Out”. Removed with its date.

assigned_room_type

5.4%

cancelled when the assigned room differs from the reserved one (vs 41.6%)

required_car_parking_spaces

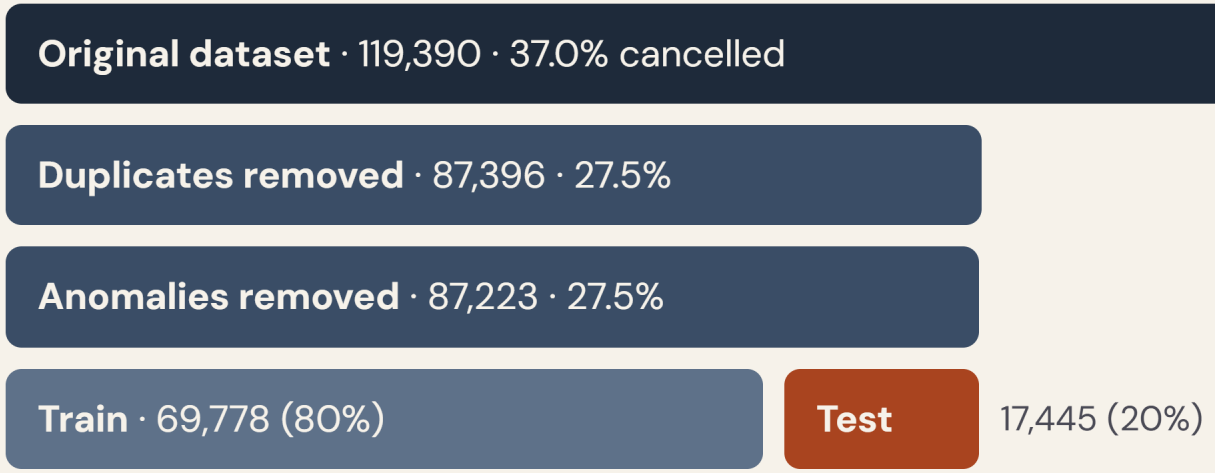
0%

cancelled when a parking space was requested

Rooms and parking are recorded **at check-in**: they exist only for guests who arrived. All four were removed, and the “reserved vs assigned room” feature of the proposal was not created.

3 · DATA PREPARATION

After cleaning, cancellations drop from 37% to 27.5%



Stratified split: 27.5% cancelled in both sets

Duplicates: no new information, and they could land in both train and test.

Missing values by meaning: no agent → "No agent"; company → flag *has_company*.

4 new features and one-hot encoding → 76 features.

4–5 · MODELING AND EVALUATION

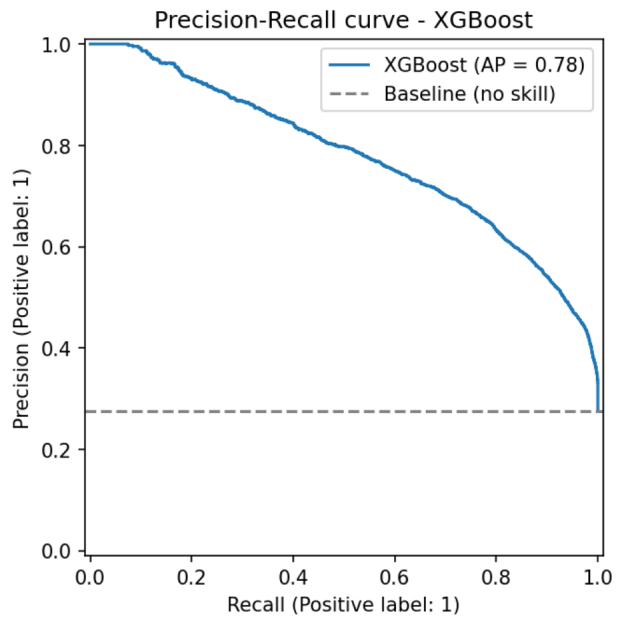
XGBoost is selected for the highest PR-AUC

Model (test set)	Precision	Recall	F1	PR-AUC
Baseline (majority class)	0.000	0.000	0.000	0.275
Logistic Regression	0.534	0.770	0.630	0.665
Decision Tree	0.555	0.826	0.664	0.718
Random Forest	0.629	0.796	0.703	0.767
XGBoost (selected)	0.612	0.821	0.701	0.778

Class weights for imbalance · tuning on the training set with stratified cross-validation · test scores close to CV, so **no overfitting**. Random Forest is very close (slightly higher F1).

5 · CHOICE OF THE THRESHOLD

A threshold of 0.4 catches 89% of cancellations

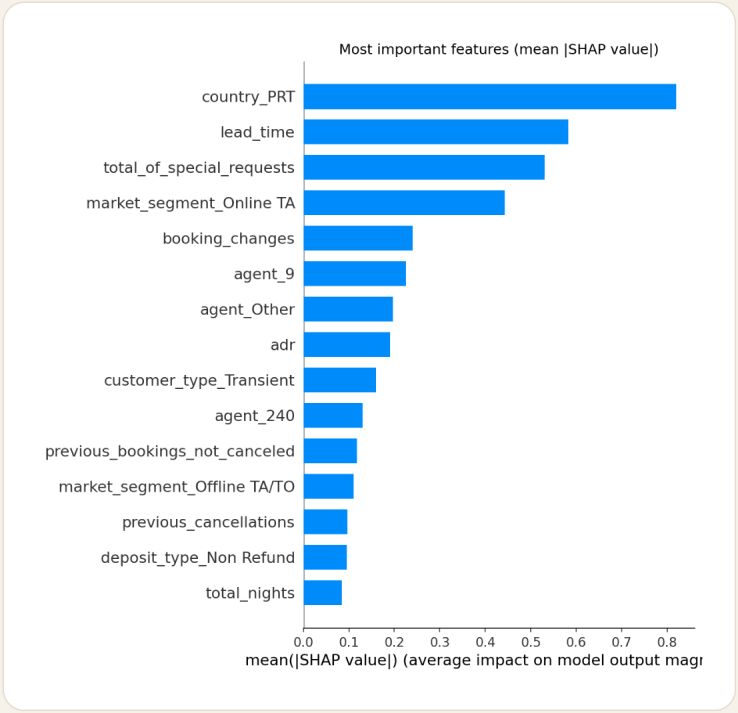


Threshold	Precision	Recall	F1
0.5 (default)	0.612	0.821	0.701
0.4 (chosen)	0.556	0.887	0.683
0.6	0.682	0.737	0.708

The threshold is a **business lever**: cheap actions → lower threshold, more cancellations caught. Riskier uses like overbooking → higher threshold.

5 · INTERPRETATION WITH SHAP

Country, lead time and special requests drive the risk



Portugal ↑ domestic guests cancel more (35.7% vs 23.7%)

Lead time ↑ longer anticipation, higher risk

Special requests ↓ a sign of commitment to the stay

Online travel agencies ↑ easy to compare and re-book

Non Refund: much less important once duplicates are removed

6 · DEPLOYMENT

24% of bookings hold 63% of the cancellations

LOW RISK · P < 0.4

5.5%

cancelled · 56% of bookings

No action: leave reliable guests alone

MEDIUM RISK · 0.4–0.7

34.8%

cancelled · 20% of bookings

Reminder before arrival and invite to personalise the stay

HIGH RISK · P ≥ 0.7

73.3%

cancelled · 24% of bookings

Partial deposit or stricter cancellation conditions

Typical high-risk booking: ~4 months ahead, online agency, no special requests. **Overbooking** only at aggregate level: sum of probabilities per date = expected cancellations.

TEMPORAL VALIDATION AND CONCLUSIONS

Behaviour changes over time: retrain regularly

Train 2015–16 → test 2017

0.778 → 0.677

PR-AUC on future bookings; cancellation rate rises from 25.0% to 31.9%. The random split is optimistic.

Data quality was decisive: duplicates and leakage changed the conclusions.

XGBoost + threshold 0.4 detects 89% of cancellations.

Next steps: periodic retraining, real cost data for the threshold, test the actions on a subset of bookings.